Continuity

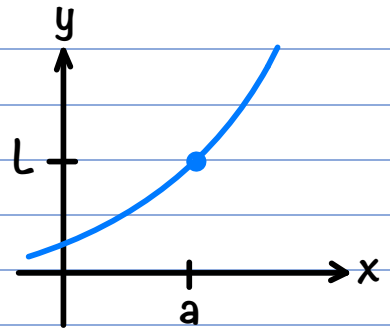
1. Continuity Definitions and Example
2. Discontinuities – Removable, Infinite, Jump and Examples
3. Continuity on an Interval and Examples
4. Intermediate Value Theorem and Examples

1. Continuity Definitions and Example

1. Continuity

A function f(x) is continuous at a number x = a if

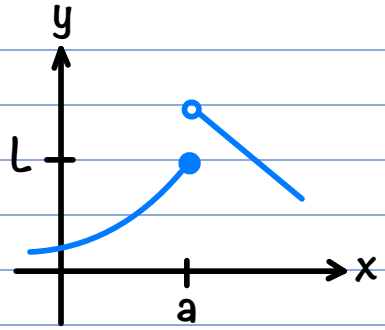
1. f(a) exists
2. $\lim_{x \to a} f(x)$ exists
3. $f(a) = \lim_{x \to a} f(x)$



2. Continuity from the Left

A function f(x) is continuous from the left if

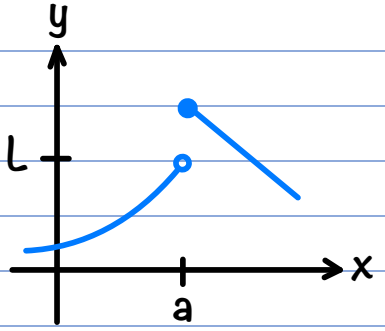
$$\lim_{x \to a^-} f(x) = f(a).$$



3. Continuity from the Right

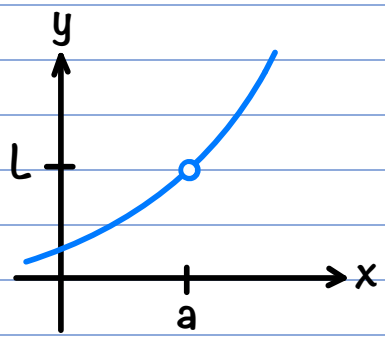
A function f(x) is continuous from the right if

$$\lim_{x \to a^+} f(x) = f(a).$$

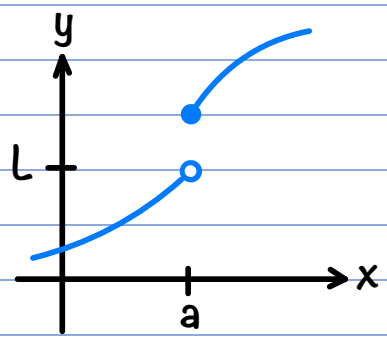


4. Discontinuity

A function f(x) is discontinuous at a number x = a if it is not continuous there.

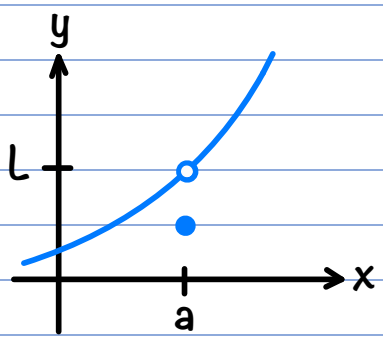


f(a) does not exist



f(a) exists

$\lim_{x \to a} f(x)$ does not exist



f(a) exists

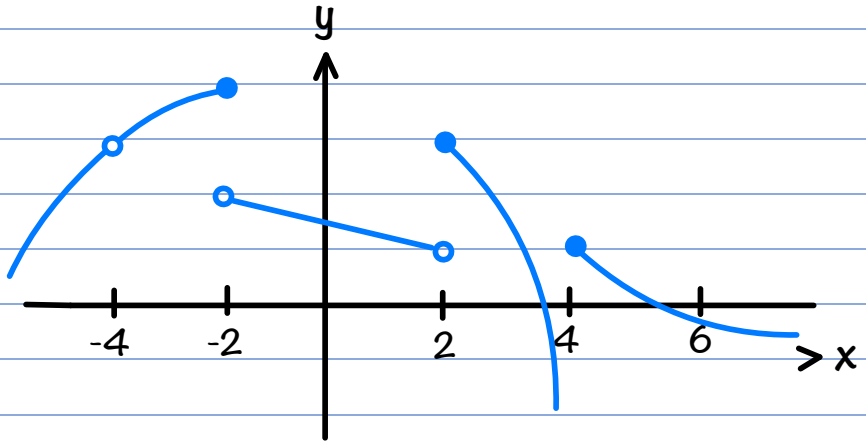
$\lim_{x \to a} f(x)$ exists

$f(a) \neq \lim_{x \to a} f(x)$

Example

(a) State the values of x at which f is discontinuous and explain why.

(b) At those numbers, determine whether f is continuous from the right, or from the left, or neither.



Interactive Animation: scalcet8 ch2 sec5 fig2

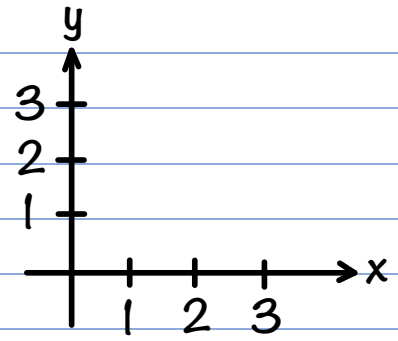
2. Discontinuities – Removable, Infinite, Jump and Examples

1. Removable Discontinuity at x = a

A removable discontinuity occurs at a point where a function's limit exists, but the function's value at that point is either undefined or doesn't equal the limit, creating a "hole" in the graph.

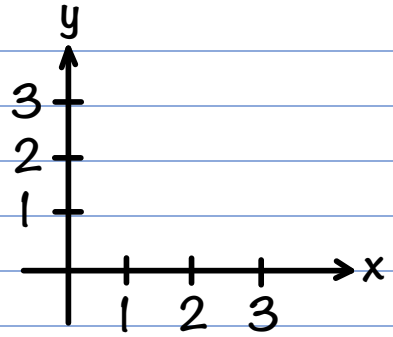
Example 1

$$f(x) = \frac{x^2 - x - 2}{x - 2}$$



Example 2

$$f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & , x \neq 2 \\ 1 & , x = 2 \end{cases}$$

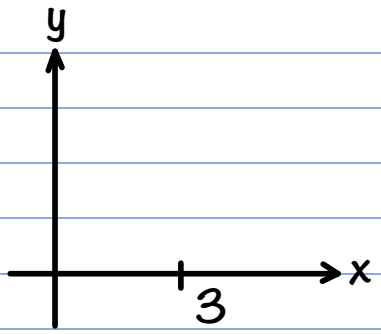


2. Infinite Discontinuity

An infinite discontinuity occurs at a point where the function's value approaches positive or negative infinity from at least one side, creating a vertical asymptote.

Example

$$f(x) = \frac{1}{(x-3)^2}$$



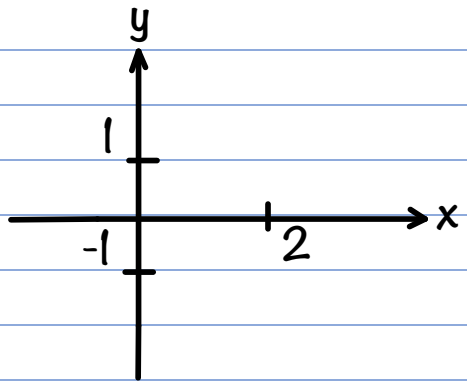
Algebraically f(x) can have a (x - a) in the denominator that can't be factored out and there is an infinite discontinuity and a vertical asymptote at x = a.

3. Jump Discontinuity

A jump discontinuity occurs when a function's one-sided limits at a specific point both exist but are not equal, causing a break in the graph where the function "jumps" from one value to another.

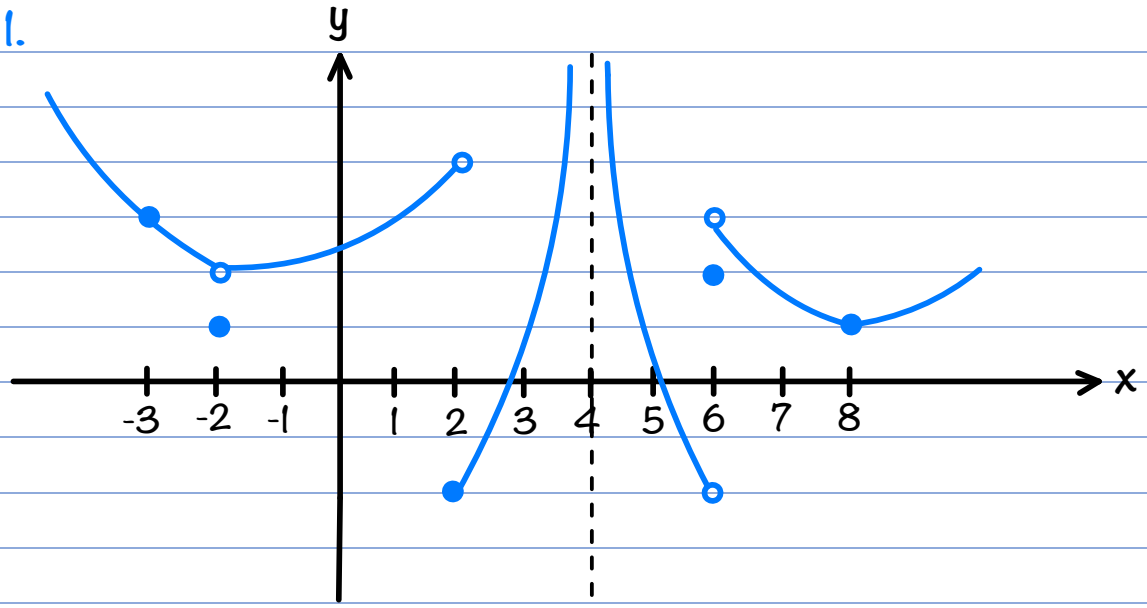
Example

$$f(x) = \frac{x-2}{|x-2|}$$

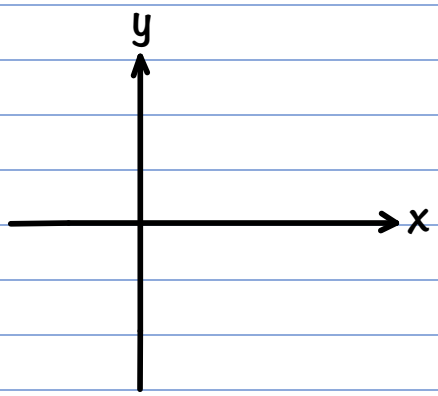


Examples - State where f(x) is discontinuous and state the type of discontinuity.

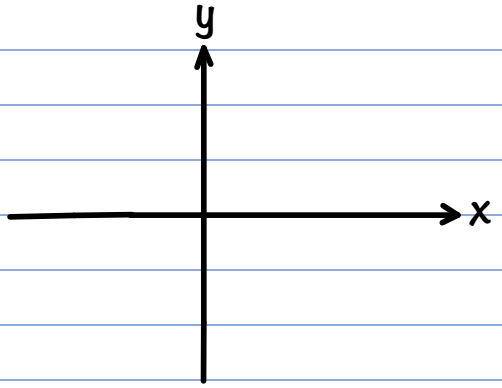
1.



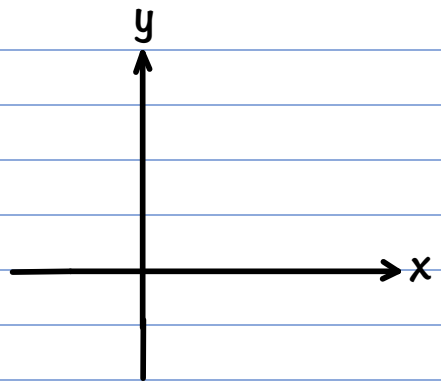
2. $f(x) = \begin{cases} \dfrac{1}{x-1} & , x \neq 1 \\ 2 & , x = 1 \end{cases}$



3. $f(x) = \begin{cases} \dfrac{x^2 - x}{x^2 - 1} & , x \neq 1 \\ 1 & , x = 1 \end{cases}$



4. $f(x) = \begin{cases} 1 + x^2 & , x < 1 \\ 4 - x & , x \geq 1 \end{cases}$



Example - Find the value of c that makes f(x) continuous everywhere.

$$f(x) = \begin{cases} cx^3 - 6 & , x < 3 \\ 4x^2 + 2cx & , x \geq 3 \end{cases}$$

3. Continuity on an Interval and Examples

The following functions are continuous at every point in their domain.

Polynomials	Exponential functions	Trigonometric functions
Rational functions	Logarithmic functions	Inverse trigonometric functions
Root functions		

A function may fail to be continuous at points where:

- The denominator equals zero
- The expression inside a square root is negative
- The argument of a logarithm is less than or equal to zero

Examples - Determine the intervals on which the function is continuous.

1. $f(x) = \dfrac{\sin x}{x + 1}$

2. $f(x) = \sqrt{x - 4}\,(x^3 + 1)$

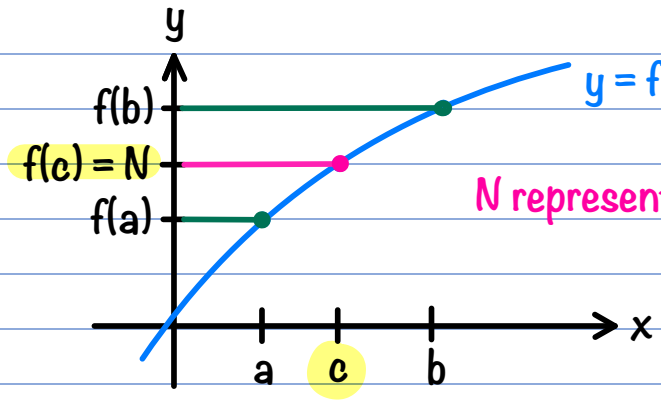
3. $f(x) = \dfrac{\ln(x - 1)}{x^2 - 4}$

4. Intermediate Value Theorem and Examples

Main idea - If you have a continuous function that goes from one value to another, it must pass through every value in between.

Intermediate Value Theorem

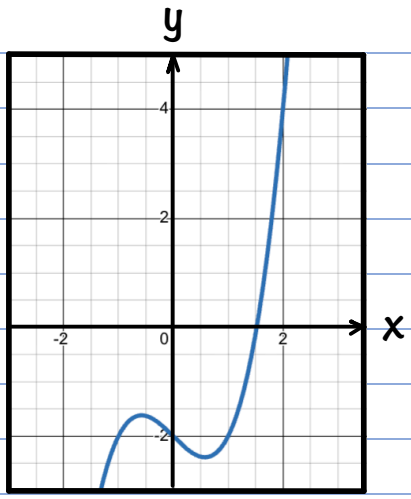
If f is continuous on the closed interval [a,b] and N is any number between f(a) and f(b) where $f(a) \neq f(b)$,
then there exists a number c in (a,b) such that $f(c) = N$.



Examples

1. Use the intermediate value theorem to show that there is a root of the given equation in the specified interval.

$x^3 - x - 2 = 0, \quad (1, 2)$



2. Show that there is a root to the equation $\sqrt{x-5} = \frac{1}{x+3}$.